sitting in our chargers—or technically speaking even EVs—could be apocalyptic for the national grid and the nation's security itself.

Another consumer concern requiring consistent attention relates to maintenance anxiety—the availability, cost and quality of customer care provided. At the heart of the issue is the crying lack of appropriately qualified and practically trained technicians to handle electronic malfunctions in EVs. To meet consumer expectations, EV makers will need to think in quite the same way the legacy original equipment manufacturers (OEMs) did—create authorised service centres and then enable those skills to get distributed through the mushrooming of a second and even third tier of appropriately skilled and equipped repair capabilities to handle the new vehicle tech. But both skill levels and instrumentation availability from the EV makers to the second and third tiers may well have to be mandated through something similar to a 'Right to Repair' law to completely address the consumer concerns. Such a law could have another very salutary effect—it would give an immediate boost to component localisation efforts and help the EV industry move from being dependent on Chinese component manufacturers to being self-reliant and world-class as the ICE, or internal combustion engine, sector has already proved to be.

The other significant determinant of future growth of the EV market could be the resale value anxiety, which in turn is directly linked to the concerns about battery replacement cost. The only capability-building that the Indian EV industry has done in the battery space relates to battery assembly—a relatively low-tech, labour-intensive operation. Imports of Advanced Chemistry Cells (ACCs), which account for almost 70 per cent of battery cost, have been the EV industry's strategy of choice. Despite India's battery demand variously estimated at between 260 GWh and 300 GWh by 2030, technology development and manufacturing capabilities in the area of cell chemistry and cell manufacturing appear to be confined to just one EV maker. As of mid-2024, there was only one accredited agency in India providing testing for battery cells. Even our 'highly acclaimed' technology research and educational institutions appear to have little or nil solid state and semi-solid state chemistry competencies or related manufacturing design capabilities. Both are crucial for reducing this dependence and ensuring long-term sustainable pricing from the consumers' standpoint. A relatively high degree of vertical integration of the 'energy delivery' component will also be crucial in EV makers' ability to provide competitively priced batteries and address consumers' anxieties. No doubt the government has put in place financial aid schemes to enable cell technology development, but how EV makers move from a trader mindset and rise to the challenge in a matter of, say, two years will ultimately determine the scheme's success.

So how would 2025 look for EVs? Each of the concerns detailed earlier would play out differently for different product/ market segments. Efforts of the industry at localisation and a constant reconfiguring of product offerings, combined with fiscal benefits, have created a wider range of price points, thus taking the E2W revolution within the grasp of a wider, more price-sensitive consumer segment. The E2W industry focus has moved to a geographic widening of the consumer base and new tech is seen and accepted as the future all over the rural hinterland too. Simultaneously, the committed entry of the legacy Indian 2W players has given EV technology a stamp of acceptance with a wider range of consumers, and the reported imminent entry of market leader Honda could well prove to be the new inflexion point. A combination of all these factors will undoubtedly help create a virtuous cycle of exponential growth. With mass-market EV motorcycles also expected to be introduced, CY '25 could well record annual sales in excess of 2.5 million units. With E3W having already proven exceptionally profitable for operators, in both the cargo and passenger segments, and efforts apace on battery swapping options, a continued volume growth of above 15 per cent seems very reasonable to expect. But it's the E4W segment that could continue to struggle. The expanding range of hybrids—with Toyota and Suzuki being joined by Honda—could continue to be the real scene-stealers. ■

> TO MEET CONSUMER EXPECTATIONS, THE INDIAN EV INDUSTRY **MUST MOVE BEYOND THE TRADER MINDSET** AND PARTICIPATE PROACTIVELY IN LOCALISATION, SPREAD OF TECHNICAL SKILLS

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